problems/MHD/smoothVortex

DynSGS and positivity setup, as coded on branch `sm/MHD` (commit `dcc3653`)

How to read the underlines. Solid underline: the setting (or the code path) is *identical in every case of `problems/MHD` to which it applies* — all nine cases for general settings, the eight `DSGS_MHD` cases for DynSGS settings, the three `:positivity_method => "conservative"` cases for the element limiter. Dashed underline: shared by a *majority* of those cases but not all of them. No underline: specific to this case or to a minority. The cross-case comparison is in `summary.pdf` of this directory.

Scope. Read off `user_inputs.jl`, the case files and the kernels they reach on `sm/MHD`; deck values are those obtained with every `JEXPRESSO_SV_*` override unset. The case is the smooth-flow accuracy test (Dao & Nazarov 2022, §5.1.1; Balsara 2004): its point is that the residual viscosity must *not* degrade a smooth high-order solution, so it is run against a plain-Galerkin twin (`JEXPRESSO_SV_VISC=none` sets `:lvisc => false`).

1 The problem as set up

Deck key / item	Value as run	Source / remark
Equations	2D ideal GLM-MHD, state $(\rho, \rho u, \rho v, E, \rho w, B_x, B_y, B_z, \psi)$	`initialize.jl`; flux code identical to Orszag–Tang's
EOS	$p = (\gamma-1)(E - \frac{1}{2}\lvert\mathbf{m}\rvert^2/\rho - \frac{1}{2}\lvert\mathbf{B}\rvert^2 - \frac{1}{2}\psi^2)$, $\gamma = 5/3$	`user_flux.jl`
Domain, mesh	$[-10, 10]^2$ ($L = 20$), doubly periodic, 16×16 quads	`vortex_L20_16x16.msh` (`JEXPRESSO_SV_L`, `_NELX`)
Discretization	CG spectral elements, LGL nodes, lumped mass; $N = 4$	`JEXPRESSO_SV_NOP`
Time integration	`CarpenterKennedy2N54`, fixed Δt, $\Delta t = 2 \times 10^{-3}$, $t_{end} = 1$	$\Delta t = 2 \times 10^{-3} \cdot 64/(n_{el}N)$; `JEXPRESSO_SV_SOLVER` can switch to Vern9, DP8, ...
Variables	`TOTAL()`, `"energy"`	
Initial data	$\rho = 1$, $\mathbf{v} = (1,1) + \kappa g\,(-y, x)$, $\mathbf{B} = \mu g\,(-y, x)$, $g = e^{(1-r^2)/2}/(2\pi)$, $p = 1 + \frac{\mu^2(1-r^2)-\kappa^2}{8\pi^2} e^{1-r^2}$, $\kappa = \mu = 1$	`sv_state`
GLM cleaning	$c_h = \max(\lvert\mathbf{v}\rvert + c_f)$ of the initial data, constant; source $S_\psi = -(c_h/0.18)\,\psi$	`initialize.jl`, `user_source.jl`
Sources	`:lsource => true` (the ψ damping only)	
Boundary conditions	periodic (the free-slip fallback of `user_bc.jl` never fires)	
Other switches	`:lfilter`, `:lkep`, `:entropy_variables`, `:lrichardson`, `:ladapt` false	

2 DynSGS: deck keys and effective values

Deck key / item	Value as run	Source / remark
`:visc_model`	`DSGS_MHD()`, `:lvisc => true`	2D element kernel `SGS.jl:1598`
`:`μ	all 1	

Deck key / item	Value as run	Source / remark
`:dsgs_sensor`	`"residual"`	`JEXPRESSO_SV_SENSOR`
`:dsgs_CR`	1.0	`JEXPRESSO_SV_CR`
`:dsgs_Cmax`	0.5	`JEXPRESSO_SV_CMAX`
`:dsgs_Cmin`	0	`JEXPRESSO_SV_CMIN`
`:dsgs_gamma`	$= \gamma_{mhd} = 5/3$	
`:dsgs_Prt`	1.0	no effect (no Prandtl factor in the conserved branch)
`:dsgs_rel`	1.0	`JEXPRESSO_SV_REL`
`:dsgs_norms`	`"domain"`	`JEXPRESSO_SV_NORMS`
`:dsgs_conserved`	true	see Section 4: the primitives are *not* conserved
`:dsgs_nazarov_energy`	false (default)	
`:dsgs_fast_floors`,	false (defaults)	
`:dsgs_thermal_residual`		
`:dsgs_ref_weight`,	false, false, false	defaults
`:dsgs_nodal_rho`,		
`:dsgs_reference`		
`:dsgs_hold_steps`	2	`JEXPRESSO_SV_HOLD`
`:dsgs_cutoff`	0	`JEXPRESSO_SV_CUTOFF`
`:dsgs_freeze_stage`	false	`JEXPRESSO_SV_FREEZE`
`:ldsgs_nodal`	false (element form)	default
`:dsgs_local_rel`	1.0 (default)	unused (domain norms)

3 DynSGS as it runs for this case

Element K, node $I = (i, j)$, $m_I^K = \omega_i \omega_j J_{K,ij}$, $\Delta_K = \Delta_K^{elem}/(N+1)$ with Δ_K^{elem} the smaller corner-to-corner extent of K: $\Delta_K^{elem} = 1.25$, $\Delta_K = 0.25$.

1. **History and hold.** Rotation once per step; two held steps ($\nu = 0$ then, since $C_{\min} = 0$).
2. **Time derivative.** Residual sensor: BDF2 on $(\mathbf{q}^n, \mathbf{q}^{n-1}, \mathbf{q}^{n-2})$ at the stage on t^n, the three-point derivative through $(\mathbf{q}(\tau), \mathbf{q}^n, \mathbf{q}^{n-1})$ at later stages (formulas as in `rhs.jl:964-971`).
3. **Residual.** $R_{K,I}^s = |D_t q_I^s - r_I^{K,s}/m_I^K|$, r^K the element's own weak inviscid RHS (flux divergence and the ψ source). The mesh is periodic, so the Dirichlet-node list is empty and nothing is zeroed.
4. **Normalization** (domain, MPI-reduced): $n^s = \max(S^s, \sigma^s) + 10^{-16}$ with the mean-state floors $\sigma^\rho = \bar{\rho}$, $\sigma^{\mathbf{m}} = \bar{\rho}\bar{c}$, $\sigma^E = \bar{\rho}\bar{c}^2$, $\sigma^{\mathbf{B}} = \sqrt{\bar{\rho}}\,\bar{c}$, $\bar{c} = \sqrt{\gamma \bar{p}/\bar{\rho}}$. Here $\rho \equiv 1$, so $S^\rho = 0$ and the continuity residual is measured against $\bar{\rho} = 1$ (this is the case for which the floor was raised from 10^{-3} of the scale to the scale itself, `:dsgs_rel`).
5. **Ratio and cap.** $\mathcal{R}_K = \max_{I \in K} \max_{s<8} R_{K,I}^s/n^s$ (ψ excluded); $w_K = \max_I(|\mathbf{v}_I| + \sqrt{\max(\gamma p_I/\rho_I + |\mathbf{B}_I|^2/\rho_I, 0)})$.
6. **Coefficient.** $\nu_K = \max(0, \min(0.5\,\Delta_K w_K,\ \Delta_K^2 \mathcal{R}_K))$, no floor, no cutoff, one value per element, recomputed at every stage.
7. **Slot coefficients.** Because `:dsgs_conserved => true`, every slot gets the kinematic ν_K (`SGS.jl:1858-1888`): no element-mean $\bar{\rho}$, no Prandtl factor.

4 The viscous operator actually assembled

`user_primitives.jl` of this case returns the *physical* variables $(\rho, u, v, T{=}p/\rho, w, B_x, B_y, B_z, \psi)$ (it is a copy of the Orszag–Tang file), while the deck selects the conserved coefficients. In `_expansion_visc!` the flag `lcons` then turns off the deviatoric stress and the $\tau \cdot \mathbf{v}$/resistive work (`rhs.jl:2959-2970`), so every slot receives a scalar flux $\nu_K \nabla$(primitive):

Equation	Flux added	Remark
ρ	$\nu_K \nabla \rho$	mass diffusion on
ρu, ρv, ρw	$\nu_K \nabla u$, $\nu_K \nabla v$, $\nu_K \nabla w$	no ρ factor, no deviatoric stress
E	$\nu_K \nabla T$, $T = p/\rho$	no $\tau \cdot \mathbf{v}$, no $\eta\, \mathbf{B} \cdot \nabla \mathbf{B}$
B_x, B_y, B_z	$\nu_K \nabla B_{x,y,z}$	resistivity $\eta = \nu$
ψ	$\nu_K \nabla \psi$	

This is neither the conserved form (which needs $\nabla(\rho \mathbf{v})$, ∇E) nor the physical form (which needs $\bar{\rho}\nu$, the stress and the work terms). All fluxes are in divergence form, so mass, momentum, total energy and $\mathbf{B}$ are still conserved; but the kinetic and magnetic energy removed by the momentum and induction terms is not returned to E. With $\rho \approx 1$ the momentum terms coincide with $\nabla \cdot (\nu \nabla(\rho \mathbf{v}))$ to $O(\rho - 1)$.

5 Positivity

- `:lpositivity` false; no stage limiter; no state modification of any kind.
- No flux-only floor either: `pressure_mhd` is the unguarded Orszag–Tang one.
- None is needed by design: the data are smooth, $\min p = 1 - 1/(8\pi^2) \approx 0.987$ initially (at $r = 1$), and the case measures accuracy, not robustness.

6 Observations: code versus comments

1. The header of `user_primitives.jl` says the momentum slots get "the full deviatoric stress built by the kernel, $\rho\nu$" and the energy "$\kappa \nabla T$ ... $+\tau \cdot u$ viscous work". That is what happens with `:dsgs_conserved => false`; with the deck's `true` none of the three is applied (Section 4). The `README.md` does not mention `:dsgs_conserved`.
2. Everything else in the deck matches its comments (domain norms, hold 2, no floor, no cutoff, the residual sensor).

Appendix: code map (branch `sm/MHD`, commit `dcc3653`)

Mechanism	**Where (file:line)**
Deck defaults of every `:dsgs_*` key	`src/io/mod_inputs.jl:1085-1376`
Deck defaults of `:lpositivity`, floors, report `:positivity_method`; rebuild of the RK solver	`src/io/mod_inputs.jl:456-477` `src/io/mod_inputs.jl:828-838`
DynSGS history rotation, stage weights, hold, freeze	`src/kernel/operators/rhs.jl:927-984`
Stencil selection `_dsgs_stencil`	`rhs.jl:141`
Hold length `_dsgs_hold_steps` ($\max(2, H) + 1$ rotations)	`rhs.jl:182`
Residual right-hand side (element / legacy / Dirichlet)	`rhs.jl:188-280`; legacy fill `rhs.jl:282`; boundary zeroing `rhs.jl:313, 346`
1D `DSGS_MHD` dispatch	`rhs.jl:1398-1470`
2D `DSGS_MHD` dispatch (sets `dsgs_ref_weight`, `dsgs_split_energy`, `dsgs_nodal_rho`)	`rhs.jl:1517-1610`
2D viscous assembly `_viscous_rhs_el_2d_dsgs!`	`rhs.jl:1796`
2D weak viscous flux `_expansion_visc!` (conserved flag, ref. weight, split energy, $\tau \cdot \mathbf{v}$, resistive work)	`rhs.jl:2943-3210` (`lcons` 2959, weight 2978, split 3050, $\tau \cdot \mathbf{v}$ 3076, $\eta\, \mathbf{B} \cdot \nabla \mathbf{B}$ 3091)
1D weak viscous flux	`rhs.jl:2823`; element loop `rhs.jl:1321`
2D `DSGS_MHD` element kernel	`src/kernel/physics/SGS.jl:1598-1912`
1D `DSGS_MHD` element kernel	`SGS.jl:1929-2120`
Nodal kernels (off in every MHD deck; BW can switch it on with `JEXPRESSO_BW_NODAL=1`)	`SGS.jl:2144` (1D), `SGS.jl:2515` (2D)
Normalization floor `_dsgs_denom`, `:dsgs_rel` multiplier	`SGS.jl:676-679`; set per call in `rhs.jl:192`
Smoothness cutoff `_dsgs_cut`	`SGS.jl:703`
`SGS_diffusion(::DSGS_MHD)` (nodal-ρ switch)	`SGS.jl:547-569`

Mechanism	**Where (file:line)**
Element length Δ_K^{elem} (2D: min of corner extents)	`src/kernel/mesh/mesh.jl:6430-6457`; 1D Δx: `mesh.jl:5863`
`uprimitive` width ($n_{eqs}+2$ in 2D, $n_{eqs}+1$ in 1D)	`src/kernel/globalStructs.jl:139-153`
Realizability repair (`"repair"`, unused by the MHD decks)	`src/kernel/positivity/Positivity.jl:294`; driver `positivity_driver.jl:124`; called in `rhs.jl:821`
Conservative element limiter (`"conservative"`)	`Positivity.jl:433-559`; driver `positivity_driver.jl:245-337`
Main `solve` call (`saveat` non-empty ⇒ `calck = true`)	`src/kernel/solvers/TimeIntegrators.jl:855`

Stage-limiter calls in `CarpenterKennedy2N54`: with `williamson_condition = false` the limiter is called before the right-hand side of stages 2–5 and once after stage 5 (OrdinaryDiffEq v6.90/v6.95, `low_storage_rk_perform_step.jl`). In OrdinaryDiffEq `calck = true` forces `williamson_condition = false` in the 2N cache, and Jexpresso's `solve` passes a non-empty `saveat`, so this holds for every deck that hands the solver a stage limiter. (Jexpresso ships no `Manifest.toml`; the package version is whatever the environment resolves, and the check above was made against v6.90/v6.95.)